

## 1.12 Summary and Notes

### 1.12.1 Summary

*Idea.*

1. Row rank equals column rank, and  $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A})$  is the number of columns.  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ , so the normal equations are always consistent (Theorem 1.2.1, Theorem 1.2.2 and Proposition 1.2.3).
2. A generalized inverse satisfies  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ . Every matrix has one, built from any nonsingular  $r \times r$  submatrix. The Moore–Penrose inverse is the unique one that is also reflexive and makes  $\mathbf{A}\mathbf{A}^+$  and  $\mathbf{A}^+\mathbf{A}$  symmetric (Definition 1.3.1, Theorem 1.3.2, Proposition 1.2.4 and Definition 1.3.2).
3.  $\mathbf{A}\mathbf{x} = \mathbf{b}$  is consistent iff  $\mathbf{A}\mathbf{A}^-\mathbf{b} = \mathbf{b}$ , and then its solutions are  $\mathbf{A}^-\mathbf{b} + (\mathbf{I} - \mathbf{A}^-\mathbf{A})\mathbf{z}$ . A linear function  $\mathbf{q}^\top \mathbf{x}$  is the same for all solutions iff  $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$  (Theorem 1.4.1 and Corollary 1.4.2).
4. Orthogonal matrices preserve lengths and angles. Gram–Schmidt gives the QR factorization, and  $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$  (Proposition 1.5.3, Theorem 1.5.2 and Proposition 1.5.1).
5.  $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A})$ . Block elimination gives the determinant and inverse of a partitioned matrix through Schur complements, and the Sherman–Morrison–Woodbury identity as a special case (Theorem 1.6.1, Theorem 1.6.3, Theorem 1.6.4 and Theorem 1.6.5).
6. A symmetric matrix is  $\mathbf{Q}\mathbf{A}\mathbf{Q}^\top$  with  $\mathbf{Q}$  orthogonal. Positive definiteness is equivalent to positive eigenvalues, to a Cholesky factorization, and to positive leading minors. Nonnegative definite matrices have a unique nonnegative definite square root (Theorem 1.7.2, Theorem 1.7.4 and Theorem 1.7.6).
7. The Rayleigh quotient  $\mathbf{x}^\top \mathbf{A}\mathbf{x} / \mathbf{x}^\top \mathbf{x}$  ranges over  $[\lambda_{\min}, \lambda_{\max}]$  (Theorem 1.7.7 and Corollary 1.7.8).
8. Every matrix is  $\mathbf{U}\mathbf{D}\mathbf{V}^\top$ . The condition number  $\kappa = \sigma_{\max}/\sigma_{\min}$  bounds error growth, is squared by forming  $\mathbf{X}^\top \mathbf{X}$ , and its smallest singular vector exhibits near-dependences among the columns (Theorem 1.8.1, Definition 1.8.1 and Proposition 1.8.2).
9. Idempotent matrices have eigenvalues 0 and 1 and rank equal to trace. Symmetric ones are  $\mathbf{Q}_1\mathbf{Q}_1^\top$ . If symmetric matrices summing to  $\mathbf{I}$  have ranks summing to  $n$ , they are idempotent and mutually orthogonal (Proposition 1.9.1, Theorem 1.9.2 and Theorem 1.9.3).
10. Kronecker products factor balanced designs.  $\text{vec}(\mathbf{A}\mathbf{B}\mathbf{C}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$  (Proposition 1.10.1).
11.  $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})/\partial \mathbf{x} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$ . Quadratic functions with nonnegative definite matrices are minimized by solving a linear system, with or without linear constraints (Theorem 1.11.1, Proposition 1.11.2 and Proposition 1.11.3).

### 1.12.2 Notes and sources

**Coverage and framing.** The selection of results follows what the rest of the book uses. Rencher and Schaalje (2008, chapter 2) was the checklist for topics: its sections on positive definite matrices, generalized inverses, orthogonal matrices, idempotent matrices and matrix

calculus correspond to [Section 1.7](#), [Section 1.3](#), [Section 1.5](#), [Section 1.9](#) and [Section 1.11](#). Several results that Rencher and Schaalje state without proof or with a reference, such as the general solution of a consistent system, the characterizations of positive definiteness and the sums of idempotent matrices, are proved here. The vector-space point of view, in which column spaces, null spaces and orthogonal complements come first and matrices second, is that of Christensen (2020, appendices A and B). Seber and Lee (2003, appendix A) is a compact list of further matrix facts. The Kronecker product results in [Section 1.10](#) are also in Christensen's appendix B.

**Linear algebra for statisticians.** Harville (1997) is the most complete treatment written for statisticians, with full proofs of the determinant facts assumed in Proposition 1.6.2 and a long development of generalized inverses. Searle (1982) is more elementary and has many worked examples from linear models. Horn and Johnson (2013) is the standard reference for eigenvalue inequalities, positive definite matrices and norms. Halmos (1958) gives the coordinate-free view of the vector-space facts used in [Section 1.2](#).

**Generalized inverses.** The four conditions of Definition 1.3.2 are due to Penrose (1955), who proved existence and uniqueness. E.H. Moore had described an equivalent inverse decades earlier. Rao (1973) made the weaker inverse  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$  the basic tool of linear-model theory, and it is his approach that [Section 1.4](#) and [Chapter 8](#) follow. The construction from a nonsingular submatrix in Theorem 1.3.2 is a matrix-free way of stating an algorithm that is often given as a sequence of steps (Searle 1982).

**Schur complements and updating formulas.** The name “Schur complement” and the notation for it vary. Zhang (2005) collects the history and many applications. The rank-one formula Eq. 1.6.7 is from Sherman and Morrison (1950), and the general identity Eq. 1.6.6 from Woodbury (1950). Henderson and Searle (1981) trace earlier appearances and compare several derivations. Obtaining the identity by comparing two expressions for one block of a partitioned inverse, as in Theorem 1.6.5, is a classical route.

**Singular values and conditioning.** Stewart (1993) tells the early history of the singular value decomposition, beginning with Beltrami and Jordan in the 1870s. The low-rank approximation theorem, Proposition 1.8.2(c) and its Frobenius form, is usually credited to Eckart and Young (1936), although Stewart points out that Schmidt had proved it earlier in the setting of integral equations. Golub and Van Loan (2013) is the reference for computing the decomposition and for perturbation bounds such as Eq. 1.8.3, and Higham (2002) for their refinements. The use of scaled condition numbers and singular vectors to diagnose collinearity is due to Belsley et al. (1980). [Chapter 26](#) returns to it.

**Kronecker products and calculus.** Henderson and Searle (1979) survey the vec operator and its uses in multivariate statistics. Magnus and Neudecker (2019) develop matrix calculus systematically through differentials, which avoid the bookkeeping of the entrywise definitions used in [Section 1.11](#) and handle symmetric matrix arguments cleanly.

### 1.12.3 References

- Belsley, David A., Kuh, Edwin and Welsch, Roy E. (1980). *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. New York: Wiley.
- Christensen, Ronald (2020). *Plane Answers to Complex Questions: The Theory of Linear Models*. 5th edition. Cham: Springer.
- Eckart, Carl and Young, Gale (1936). The Approximation of One Matrix by Another of Lower Rank. *Psychometrika* 1(3), 211–218.
- Golub, Gene H. and Van Loan, Charles F. (2013). *Matrix Computations*. 4th edition. Baltimore: Johns Hopkins University Press.

- Halmos, Paul R. (1958). *Finite-Dimensional Vector Spaces*. 2nd edition. Princeton, NJ: Van Nostrand.
- Harville, David A. (1997). *Matrix Algebra From a Statistician's Perspective*. New York: Springer.
- Henderson, Harold V. and Searle, Shayle R. (1979). Vec and Vech Operators for Matrices, with Some Uses in Jacobians and Multivariate Statistics. *Canadian Journal of Statistics* 7(1), 65–81.
- Henderson, Harold V. and Searle, Shayle R. (1981). On Deriving the Inverse of a Sum of Matrices. *SIAM Review* 23(1), 53–60.
- Higham, Nicholas J. (2002). *Accuracy and Stability of Numerical Algorithms*. 2nd edition. Philadelphia: SIAM.
- Horn, Roger A. and Johnson, Charles R. (2013). *Matrix Analysis*. 2nd edition. Cambridge: Cambridge University Press.
- Magnus, Jan R. and Neudecker, Heinz (2019). *Matrix Differential Calculus with Applications in Statistics and Econometrics*. 3rd edition. Hoboken, NJ: Wiley.
- Penrose, Roger (1955). A Generalized Inverse for Matrices. *Mathematical Proceedings of the Cambridge Philosophical Society* 51(3), 406–413.
- Rao, C. Radhakrishna (1973). *Linear Statistical Inference and Its Applications*. 2nd edition. New York: Wiley.
- Rencher, Alvin C. and Schaalje, G. Bruce (2008). *Linear Models in Statistics*. 2nd edition. Hoboken, NJ: Wiley.
- Searle, Shayle R. (1982). *Matrix Algebra Useful for Statistics*. New York: Wiley.
- Seber, George A. F. and Lee, Alan J. (2003). *Linear Regression Analysis*. 2nd edition. Hoboken, NJ: Wiley.
- Sherman, Jack and Morrison, Winifred J. (1950). Adjustment of an Inverse Matrix Corresponding to a Change in One Element of a Given Matrix. *The Annals of Mathematical Statistics* 21(1), 124–127.
- Stewart, Gilbert W. (1993). On the Early History of the Singular Value Decomposition. *SIAM Review* 35(4), 551–566.
- Woodbury, Max A. (1950). *Inverting Modified Matrices*. Princeton, NJ: .
- Zhang, Fuzhen (2005). *The Schur Complement and Its Applications*. New York: Springer.